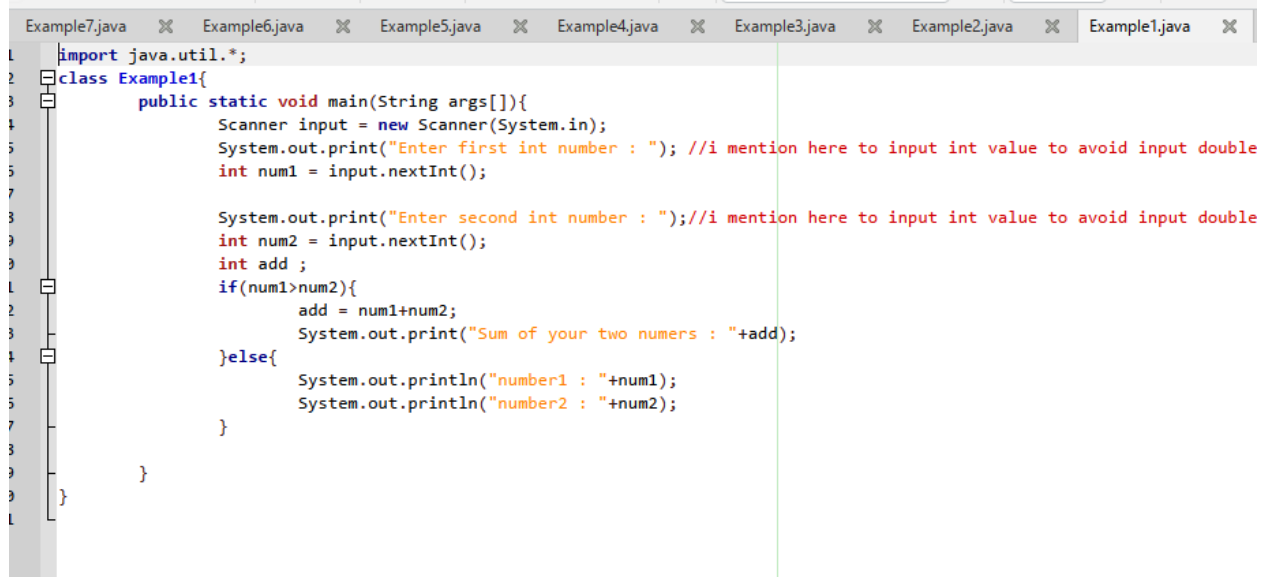
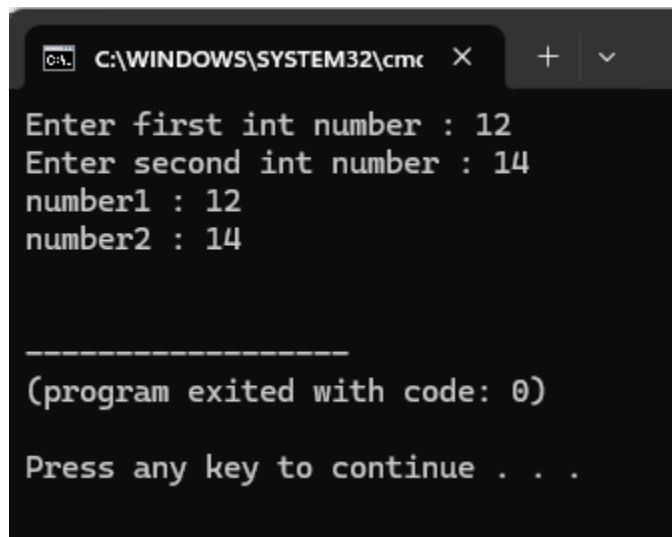


Q1.



```
1 import java.util.*;
2 class Example1{
3     public static void main(String args[]){
4         Scanner input = new Scanner(System.in);
5         System.out.print("Enter first int number : "); //i mention here to input int value to avoid input double
6         int num1 = input.nextInt();
7
8         System.out.print("Enter second int number : "); //i mention here to input int value to avoid input double
9         int num2 = input.nextInt();
10        int add ;
11        if(num1>num2){
12            add = num1+num2;
13            System.out.print("Sum of your two numbers : "+add);
14        }else{
15            System.out.println("number1 : "+num1);
16            System.out.println("number2 : "+num2);
17        }
18    }
19 }
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter first int number : 12
Enter second int number : 14
number1 : 12
number2 : 14

-----
(program exited with code: 0)
Press any key to continue . . .
```

Q2.

```
Example0.java  Example1.java  Example0.java  Example0.java  Example0.java  Example0.java
import java.util.*;
class Example2{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter any integer number : ");
        int num = input.nextInt();
        int absolute;
        if(num<0){
            absolute = -num;
            System.out.print("absolute : "+absolute);
        }else{
            absolute = num;
            System.out.print("absolute : "+absolute);
        }
    }
}
```

Outputs.

```
C:\WINDOWS\SYSTEM32\cmd  X  +  v
Enter any integer number : -156
absolute : 156
-----
(program exited with code: 0)
```

```
C:\WINDOWS\SYSTEM32\cmd  X  +  v
Enter any integer number : 134
absolute : 134
-----
(program exited with code: 0)
```

Q3.

```
Example9.java  Example8.java  Example7.java  Example6.java  Example5.java
import java.util.*;
class Example3{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter your Chemistry marks : ");
        double chem = input.nextDouble();

        System.out.print("Enter your Physics marks : ");
        double phy = input.nextDouble();

        System.out.print("Enter your Combined maths marks : ");
        double maths = input.nextDouble();

        double tot = chem+phy+maths;
        double avg = tot/3;

        System.out.println("total is : "+ tot);
        System.out.println("avarege is : "+avg);

        if(avg<=75){
            System.out.print("Fail");
        }else{
            System.out.print("Pass");
        }
    }
}
```

Outputs.

```
Enter your Chemistry marks : 27
Enter your Physics marks : 47
Enter your Combined maths marks : 59
total is : 133.0
avarege is : 44.333333333333336
Fail
```

```
Enter your Chemistry marks : 79
Enter your Physics marks : 87
Enter your Combined maths marks : 99
total is : 265.0
avarege is : 88.33333333333333
Pass
```

Q4.

```
Example10.java  X  Example9.java  X  Example8.java  X  Example7.java  X  Example6.java  X  E
import java.util.*;
class Example4{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the unit price : ");
        double upri = input.nextDouble();

        System.out.print("Enter the Quantity that you bought : ");
        int qua = input.nextInt();

        double tot = upri*qua;
        System.out.println("Total price : "+ tot);

        if (tot>1500){
            System.out.print("You are entitled to the super draw");
        }else{
            System.out.print("try again");
        }
    }
}
```

Outputs.

```
Enter the unit price : 140
Enter the Quantity that you bought : 40
Total price : 5600.0
You are entitled to the super draw
```

```
Enter the unit price : 154
Enter the Quantity that you bought : 5
Total price : 770.0
try again
```

Q5.

```
import java.util.*;
class Example5{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the unite price : ");
        double upri = input.nextDouble();

        System.out.print("Enter the Quantity : ");
        int qua = input.nextInt();

        double tot = upri*qua;
        System.out.println("your price is "+tot);
        double dis;
        double nTot;
        if(tot>500){
            dis = tot*0.05;
            nTot = tot - dis;
            System.out.printf("New price with discount : %.2f \n ",nTot);

        }else{
            System.out.print("No discount given");
        }
    }
}
```

Outputs.

```
C:\WINDOWS\system32\cmd - X
Enter the unite price : 246
Enter the Quantity : 13
your price is 3198.0
New price with discount : 3038.10

=====
(program exited with code: 0)
```

```
Enter the unite price : 132
Enter the Quantity : 3
your price is 396.0
No discount given
```

Q6.

```
ample12.java  X Example11.java  X Example10.java  X Example9.java  X Example8.java  X E
import java.util.*;
class Example6{
    public static void main(String args[]){
        Scanner input =new Scanner(System.in);
        System.out.print("Enter a Year : ");
        int year = input.nextInt();

        boolean isLeapYear = false;

        if ((year %4 ==0 && year % 100 != 0) || (year % 400 ==100)){
            isLeapYear = true;
        }

        if(isLeapYear){
            System.out.println(year + " is a leaap Year.");
        }else{
            System.out.println(year +" is not aleap Year.");
        }
    }
}
```

Outputs.

```
Enter a Year : 2004
2004 is a leaap Year.
```

```
Enter a Year : 2005
2005 is not aleap Year.

-----
(program exited with code: 0)
```

Q7.

```
import java.util.*;
class Example7{
    public static void main(String args[]){
        Scanner input =new Scanner(System.in);
        System.out.print("Enter your first integer number : ");
        int num1 = input.nextInt();

        System.out.print("Enter your Second integer number : ");
        int num2 = input.nextInt();

        System.out.print("Enter your Third integer number : ");
        int num3 = input.nextInt();

        int maxNum =0;

        if (num1>maxNum){
            maxNum = num1;
        }
        if(num2>maxNum){
            maxNum = num2;
        }
        if(num3>maxNum){
            maxNum = num3;
        }

        System.out.print("Max number is "+ maxNum);
    }
}
```

Output.

```
Enter your first integer number : 246
Enter your Second integer number : 542
Enter your Third integer number : 345
Max number is 542
```

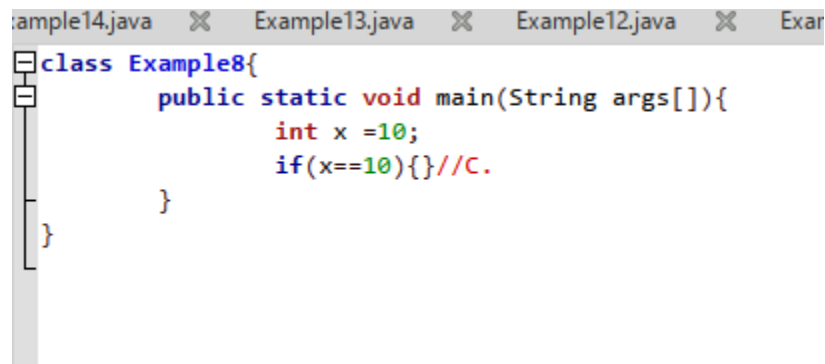
Q8.

08. Which of the following lines can be legally inserted at line 10?

```
class Example{  
    public static void main(String[] args) {  
        int x = 10;  
        //Insert code here//Line 10  
    }  
}
```

- | | |
|----------------------|--------------------------|
| A. if(x){} | B. if(x=10){} |
| C. if(x==10){} | D. if(x=100!=10){} |
| E. if((x=100)!=10){} | F. if((x=100)>0==true){} |

Output.



```
Example14.java  Example13.java  Example12.java  Exar  
class Example8{  
    public static void main(String args[]){  
        int x =10;  
        if(x==10){} //C.  
    }  
}
```

Answer. - C.

Q9.

09. Which of the following lines can be legally inserted at line 10?

```
class Example{  
    public static void main(String[] args) {  
        int x = 10;  
        boolean b=true;  
        //Insert code here//Line 10  
    }  
}
```

- | | |
|---------------------------|--------------------------|
| A. if(b){} | B. if(b=false){} |
| C. if(b==false){} | D. if(b=false==false){} |
| E. if((b=false)==false){} | F. if(b=(false==true)){} |

Answers>>

A -if(b){}

B -if(b=flase){}

C-if (b==false){}

E-if((b==false)==false){}

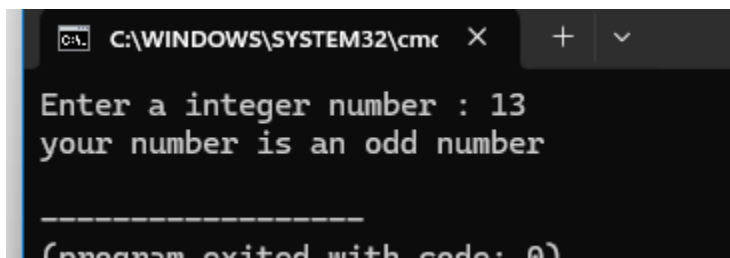
F-if(b=(false==true)){}

Q10.

```
import java.util.*;
class Example10{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a integer number : ");
        int num = input.nextInt();

        if(num%2==0){
            System.out.print("your number is an even number");
        }else{
            System.out.print("your number is an odd number");
        }
    }
}
```

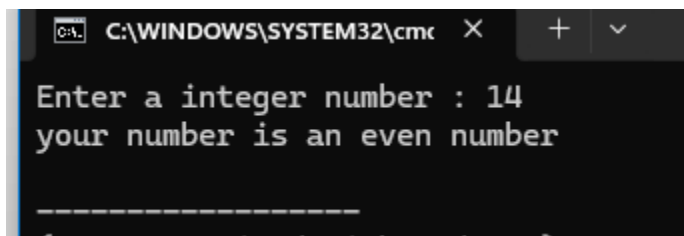
Outputs.



C:\WINDOWS\SYSTEM32\cmd X + v

Enter a integer number : 13
your number is an odd number

(program exited with code: 0)



C:\WINDOWS\SYSTEM32\cmd X + v

Enter a integer number : 14
your number is an even number

(program exited with code: 0)

Q11.

```
class Example11{  
    public static void main(String args[]){  
        boolean b=true;  
        System.out.println(b);//A  
        System.out.println(b=true);//B  
        System.out.println(b==true);//C  
        System.out.println(b!=true);//D  
        System.out.println(b=true==true);//E  
        System.out.println((b=true)==false);//F  
        System.out.println(b=(true!=false));//G  
    }  
}
```

Output.

```
true  
true  
true  
false  
true  
false  
true  
  
-----  
(program exited with code: 0)  
  
Press any key to continue . . . |
```

Q12.

```
class Example12{
    public static void main(String args[]){
        int x = 10;
        System.out.println(x=9);//line1
        System.out.println(x==9);//line2
        System.out.println(x=9!=10);//line3
        System.out.println((x=9)==10);//line4
        System.out.println((x=9)<=10);//line5
    }
}
```

javac "Example12.java" (in directory: D:\ICD124\DAY 4\N\Assignment)
Example12.java:6: error: incompatible types: boolean cannot be converted to int
System.out.println(x=9!=10);//line3
1 error
Compilation failed.

Without line3-----

```
class Example12{
    public static void main(String args[]){
        int x = 10;
        System.out.println(x=9);//line1
        System.out.println(x==9);//line2
        //System.out.println(x=9!=10);//line3
        System.out.println((x=9)==10);//line4
        System.out.println((x=9)<=10);//line5
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
9
true
false
true
```

Q13.

```
class Example13{
    public static void main(String args[]){
        int x=20,y=60;
        boolean bool;
        System.out.println(x=10);//line1
        System.out.println(bool=true);//line2
        System.out.println(x=10>0);//line3
        System.out.println((x=10)>10);//line4
        System.out.println(bool=(x=10)>0);//line5
        System.out.println(bool=x+y>100);//line6
    }
}
```

```
javac "Example13.java" (in directory: D:\ICD124\DAY 4\N\Assignment)
Example13.java:7: error: incompatible types: boolean cannot be converted to int
        System.out.println(x=10>0);//line3
                        ^
1 error
Compilation failed.
```

without line3-----

```
class Example13{
    public static void main(String args[]){
        int x=20,y=60;
        boolean bool;
        System.out.println(x=10);//line1
        System.out.println(bool=true);//line2
        //System.out.println(x=10>0);//line3
        System.out.println((x=10)>10);//line4
        System.out.println(bool=(x=10)>0);//line5
        System.out.println(bool=x+y>100);//line6
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
10
true
false
true
false
-----
```

Q14.

```
Example20.java Example19.java Example18.java Example17.java Example14.java
class Example14{
    public static void main(String args[]){
        byte b = 10;
        short s = 100;
        int x = 125;
        long l = 15000;
        float f = 1.5f;
        double d = 21.231;
        char c = 'c';
        boolean bool = 10>9;
        System.out.println(b+s+x+f+d+c+bool); //Line 1
        System.out.println(" "+b+s+x+f+d+c+bool); //Line 2
        System.out.println(b+s+x+f+d+c+" "+bool); //Line 3
        System.out.println(b+s+x+f+d+c+bool+""); //Line 4
        System.out.println(bool+b+f+d+c+" "+x+l); //Line 5
    }
}
```

```
javac "Example14.java" (in directory: D:\ICD124\DAY 4\N\Assignment)
Example14.java:14: error: bad operand types for binary operator '+'
        System.out.println(b+s+x+f+d+c+bool+""); //Line 4
                        ^
first type: double
second type: boolean
Example14.java:15: error: bad operand types for binary operator '+'
        System.out.println(bool+b+f+d+c+" "+x+l); //Line 5
                        ^
first type: boolean
second type: byte
2 errors
Compilation failed.
```

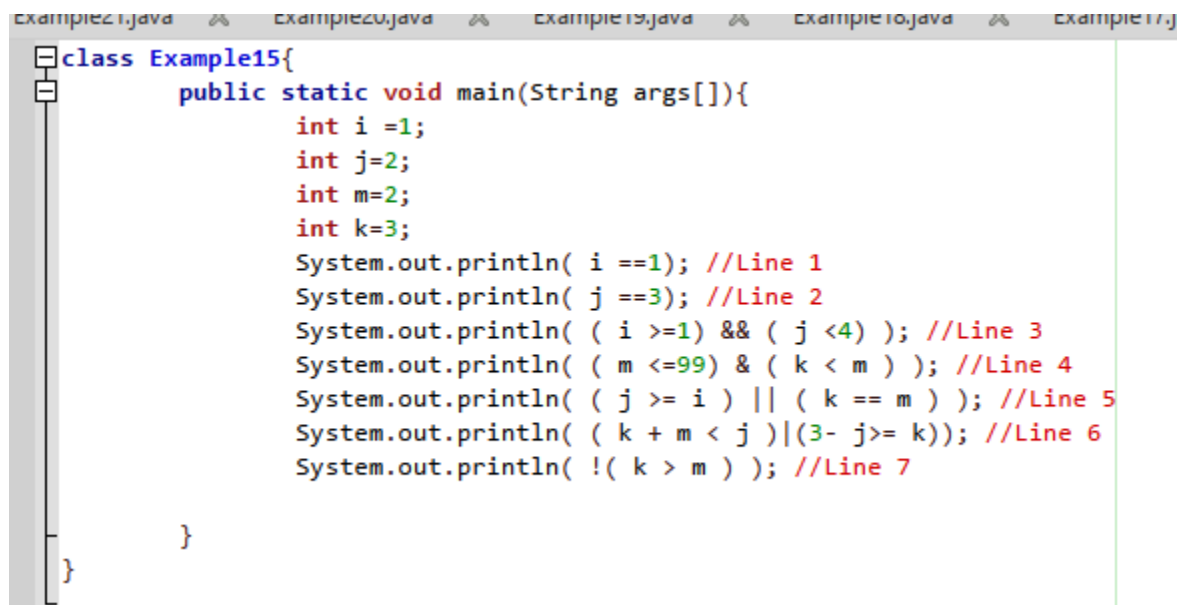
Without line4 and line5-----

```
Example19.java Example18.java Example17.java Example14.java
class Example14{
    public static void main(String args[]){
        byte b = 10;
        short s = 100;
        int x = 125;
        long l = 15000;
        float f = 1.5f;
        double d = 21.231;
        char c = 'c';
        boolean bool = 10>9;
        System.out.println(b+s+x+f+d+c+bool); //Line 1
        System.out.println(" "+b+s+x+f+d+c+bool); //Line 2
        System.out.println(b+s+x+f+d+c+" "+bool); //Line 3
        //System.out.println(b+s+x+f+d+c+bool+""); //Line 4
        //System.out.println(bool+b+f+d+c+" "+x+l); //Line 5
    }
}
```

Output.

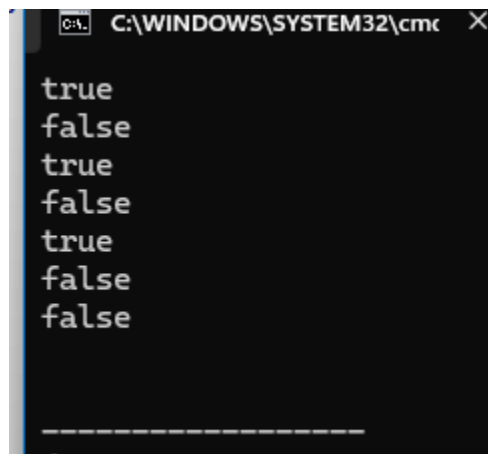
```
C:\WINDOWS\SYSTEM32\cmd
2351.521.231true
101001251.521.231true
356.731true
-----
(program exited with code: 0)
```

Q15.



```
class Example15{
    public static void main(String args[]){
        int i =1;
        int j=2;
        int m=2;
        int k=3;
        System.out.println( i ==1); //Line 1
        System.out.println( j ==3); //Line 2
        System.out.println( ( i >=1) && ( j <4) ); //Line 3
        System.out.println( ( m <=99) & ( k < m ) ); //Line 4
        System.out.println( ( j >= i ) || ( k == m ) ); //Line 5
        System.out.println( ( k + m < j ) || (3- j >= k)); //Line 6
        System.out.println( !( k > m ) ); //Line 7
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd
true
false
true
false
true
false
false
```

Q16.

```
class Example16{
    public static void main(String args[]){
        int x=0; //line 5
        switch(x){
            default: System.out.print("4 ");
            case 1 : System.out.print("1 ");
            case 2 : System.out.print("2 ");
            case 3 : System.out.print("3 ");
        }
    }
}
```

Outputs.

(A.) *int x=1;* ----

```
1 2 3
-----
(program exited with code: 0)
```

(B.) *int x=2;* ----

```
2 3
-----
(program exited with code: 0)
```

(C.) *int x=3;*-----

```
3
-----
(program exited with code: 0)
```

(D.) *int x=4;*-----

```
4 1 2 3
-----
(program exited with code: 0)
```

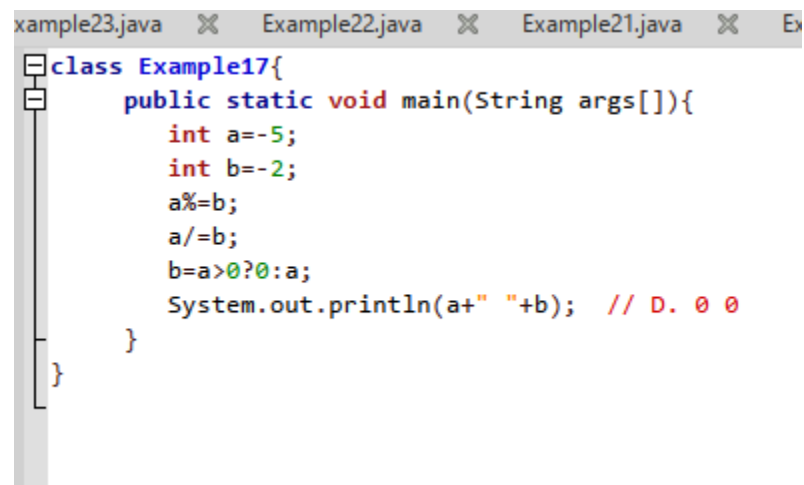
(E.) *int x=0;*-----

```
4 1 2 3
-----
(program exited with code: 0)
```

(F.) *int x=5;*-----

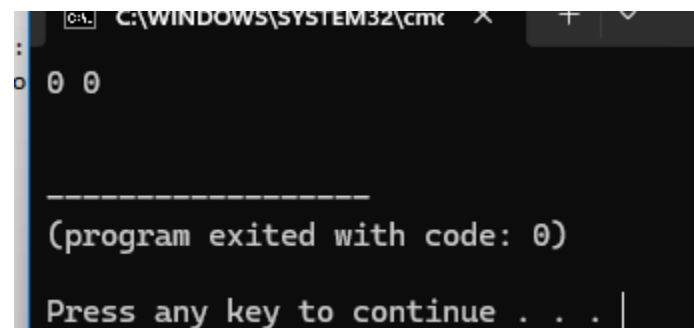
```
4 1 2 3
-----
(program exited with code: 0)
```

Q17.



```
class Example17{
    public static void main(String args[]){
        int a=-5;
        int b=-2;
        a%=b;
        a/=b;
        b=a>0?0:a;
        System.out.println(a+" "+b); // D. 0 0
    }
}
```

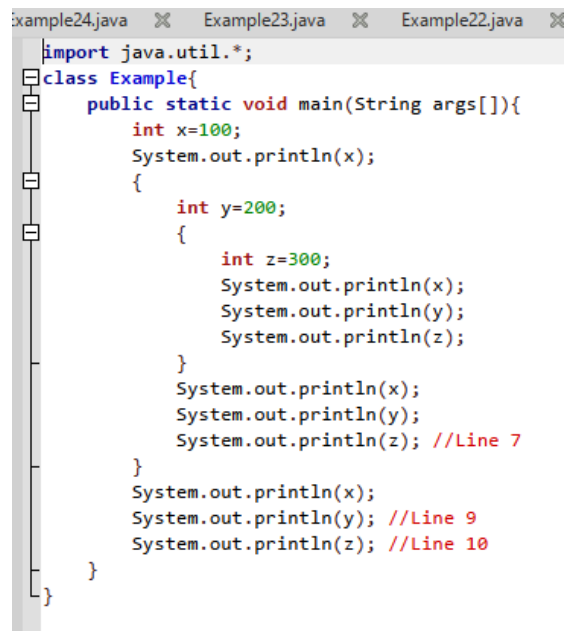
Output.



```
C:\WINDOWS\SYSTEM32\cmd
:
0 0

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q18.



```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x=100;
        System.out.println(x);
        {
            int y=200;
            {
                int z=300;
                System.out.println(x);
                System.out.println(y);
                System.out.println(z);
            }
            System.out.println(x);
            System.out.println(y);
            System.out.println(z); //Line 7
        }
        System.out.println(x);
        System.out.println(y); //Line 9
        System.out.println(z); //Line 10
    }
}
```

Answers are: A: line 1,

B: line2 ,

C:line3,

D:line 4,

E:line5 ,

F: line 6,

H: line8

Q19.

19. Given:

```
class Example{
    public static void main(String args[]){
        //Line 5
        switch(x){
            default : System.out.print("Wrong ");break;
            case 2 : System.out.print("2 ");
            case 3 : System.out.print("3 ");
            case 1 : System.out.print("1 ");break;
        }
    }
}
```

What will be the outputs when you insert the following codes at line 5?

- | | | |
|-------------|-------------|-------------|
| A. int x=1; | B. int x=2; | C. int x=3; |
| D. int x=4; | E. int x=0; | F. int x=5; |

Output.

- A. int x=1; ----- 1
- B. int x=2; ----- 2 3 1
- C. int x=3; ----- 3 1
- D. int x=4; ----- Wrong
- E. int x=0; ----- Wrong
- F. int x=5; ----- Wrong

Q20.

```
mpie20.java  Example20.java  Example24.java  Example25.java  Example22.java
import java.util.*;
class Example20{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter an any Integer number : ");
        int num = input.nextInt();

        if(num>0){
            System.out.print("Your number is a possitive number.");
        }else if(num==0){
            System.out.print("Your Number is a Zero.");
        }else{
            System.out.print("your number is negative number.");
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  X  +  v
Enter an any Integer number : 5
Your number is a possitive number.
_____
```

```
Enter an any Integer number : -12
your number is negative number.
```

```
C:\WINDOWS\SYSTEM32\cmd  X  +  v
Enter an any Integer number : 0
Your Number is a Zero.
_____
```

Q21.

```
import java.util.*;
class Example21{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the integer number1 : ");
        int num1 = input.nextInt();

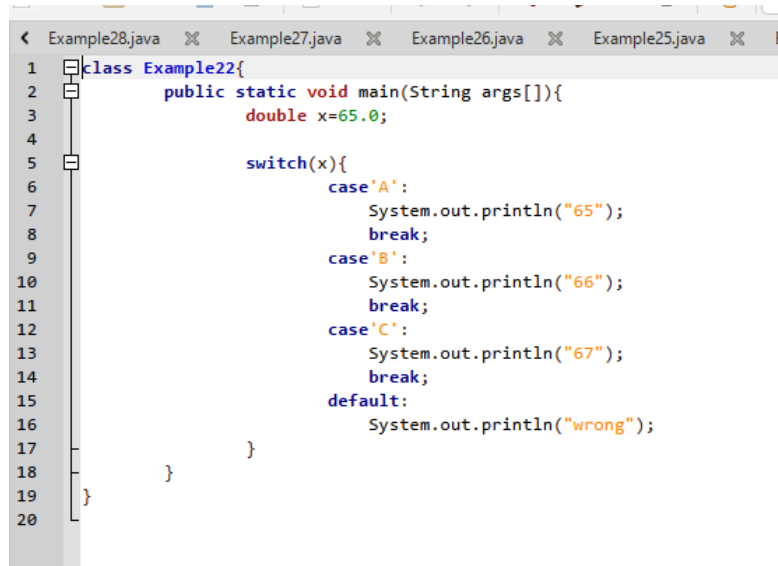
        System.out.print("Enter the integer number2 : ");
        int num2 = input.nextInt();
        int absD;
        if(num1>num2){
            absD = num1-num2;

        }else{
            absD = num2-num1;
        }
        System.out.print("Absolute difference is : " +absD);
    }
}
```

Output.

```
Enter the integer number1 : 23
Enter the integer number2 : -26
Absolute difference is :49
_____
```

Q22.



```
1 class Example22{
2     public static void main(String args[]){
3         double x=65.0;
4
5         switch(x){
6             case 'A':
7                 System.out.println("65");
8                 break;
9             case 'B':
10                System.out.println("66");
11                break;
12             case 'C':
13                System.out.println("67");
14                break;
15             default:
16                System.out.println("wrong");
17         }
18     }
19 }
20 }
```

Answer: A,B,C,D,E

Q23.

```
import java.util.*;
class Example23{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the integer Number 1 : ");
        int num1 = input.nextInt();

        System.out.print("Enter the integer Number 2 : ");
        int num2 = input.nextInt();

        if(num1<num2){
            System.out.print("The first number is less than the second number");
        }else if(num2<num1){
            System.out.print("The first number is greater than the second number");
        }else{
            System.out.print("Both are equal ");
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
Enter the integer Number 1 : 84
Enter the integer Number 2 : 37
The first number is greater than the second number
-----
```

```
Enter the integer Number 1 : 37
Enter the integer Number 2 : 59
The first number is less than the second number
-----
```

```
Enter the integer Number 1 : 33
Enter the integer Number 2 : 33
Both are equal
```

Q24.

```
import java.util.*;
class Example24{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter 1 integer number : ");
        int num1 = input.nextInt();

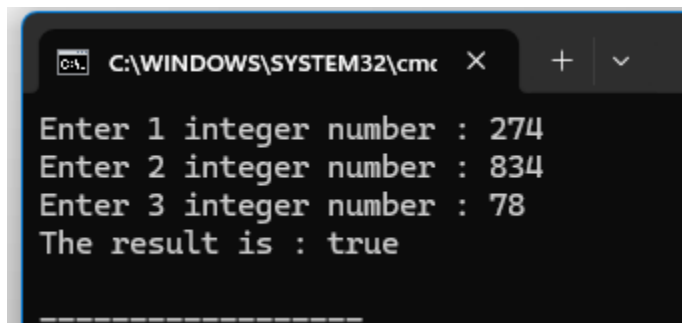
        System.out.print("Enter 2 integer number : ");
        int num2 = input.nextInt();

        System.out.print("Enter 3 integer number : ");
        int num3 = input.nextInt();

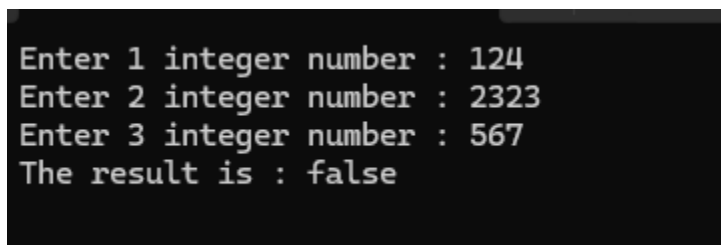
        int lastDig1 = num1%10;
        int lastDig2 = num2%10;
        int lastDig3 = num3%10;

        boolean result = (lastDig1==lastDig2) || (lastDig2==lastDig3) || (lastDig1==lastDig3);
        System.out.print("The result is : " +result);
    }
}
```

Outputs.



```
C:\WINDOWS\SYSTEM32\cmd
Enter 1 integer number : 274
Enter 2 integer number : 834
Enter 3 integer number : 78
The result is : true
```



```
Enter 1 integer number : 124
Enter 2 integer number : 2323
Enter 3 integer number : 567
The result is : false
```

Q25.

Q26.

```
import java.util.*;
class Example26{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the selling price of the product : ");
        double sprice = input.nextDouble();

        System.out.print("Enter the cost of the product : ");
        double cprice = input.nextDouble();

        if(sprice>cprice){
            System.out.print("Profit");
        }else if(cprice>sprice){
            System.out.print("Loss");
        }else{
            System.out.print("No Profit No Loss");
        }
    }
}
```

Outputs.

```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter the selling price of the product : 12000
Enter the cost of the product : 384
Profit
-----
(program exited with code: 0)
```

```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter the selling price of the product : 4000
Enter the cost of the product : 5000
Loss
-----
```

```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter the selling price of the product : 2222
Enter the cost of the product : 2222
No Profit No Loss
-----
```

Q27.

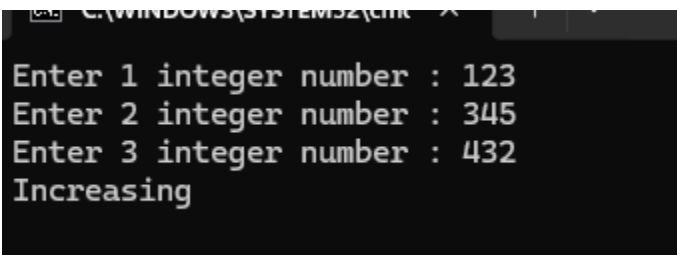
```
import java.util.*;
class Example27{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter integer number1 : ");
        double num1 = input.nextInt();

        System.out.print("Enter integer number2 : ");
        double num2 = input.nextInt();

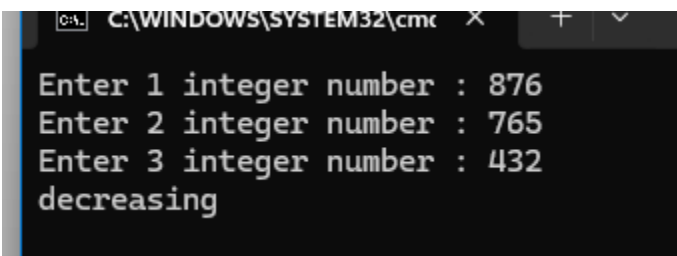
        System.out.print("Enter integer number3 : ");
        double num3 = input.nextInt();

        if(num1>num2 && num2>num3){
            System.out.print("Decreasing");
        }else if(num3>num2 && num2>num1){
            System.out.print("Increasing");
        }else{
            System.out.print("Neither increasing nor decreasing order");
        }
    }
}
```

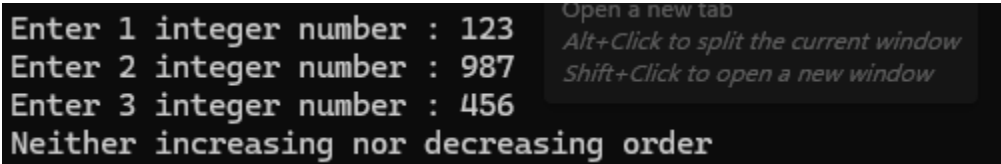
Outputs.



```
C:\WINDOWS\SYSTEM32\cmd
Enter 1 integer number : 123
Enter 2 integer number : 345
Enter 3 integer number : 432
Increasing
```



```
C:\WINDOWS\SYSTEM32\cmd
Enter 1 integer number : 876
Enter 2 integer number : 765
Enter 3 integer number : 432
decreasing
```



```
Enter 1 integer number : 123
Enter 2 integer number : 987
Enter 3 integer number : 456
Neither increasing nor decreasing order
```

Open a new tab
Alt+Click to split the current window
Shift+Click to open a new window

Q28.

```
import java.util.*;
class Example28{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter your weigh : ");
        double w = input.nextDouble();

        System.out.print("Enter your age : ");
        int a = input.nextInt();

        if(w>=50 && a>18){
            System.out.print("you can donate your blood");
        }else{
            System.out.print("you can't donate your blood");
        }
    }
}
```

Outputs.

```
Enter your weigh : 70
Enter your age : 23
you can donate your blood
-----
```

```
Enter your weigh : 60
Enter your age : 12
you can't donate your blood
-----
```

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter your weigh : 45
Enter your age : 18
you can't donate your blood
-----
(program exited with code: 0)
```

```
Enter your weigh : 45
Enter your age : 13
you can't donate your blood
-----
```

Q29.

```
Example33.java ✕ Example32.java ✕ Example31.java ✕ Example30.java ✕ Example29
import java.util.*;
class Example29{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter an integer number 1 : ");
        int num1 = input.nextInt();

        System.out.print("Enter an integer number 2 : ");
        int num2 = input.nextInt();

        if ((num1>0 && num2>0) || (num1<0 && num2<0)){
            System.out.print("True");
        }else{
            System.out.print("False");
        }
    }
}
```

Outputs.

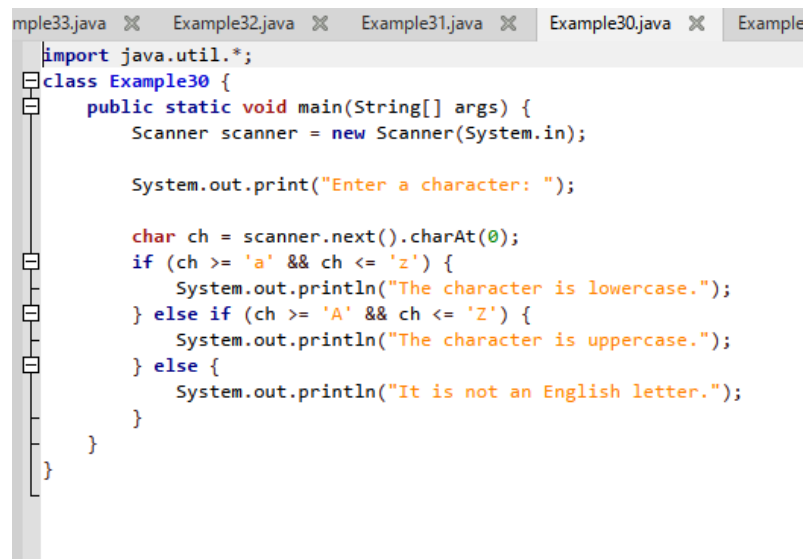
```
C:\WINDOWS\system32\cmd
Enter an integer number 1 : 23
Enter an integer number 2 : 34
True
-----
(program exited with code: 0)
```

```
C:\WINDOWS\system32\cmd
Enter an integer number 1 : -13
Enter an integer number 2 : 56
False
-----
```

```
Enter an integer number 1 : -13
Enter an integer number 2 : -14
True
```

```
Enter an integer number 1 : 14
Enter an integer number 2 : -43
False
-----
```

Q30.



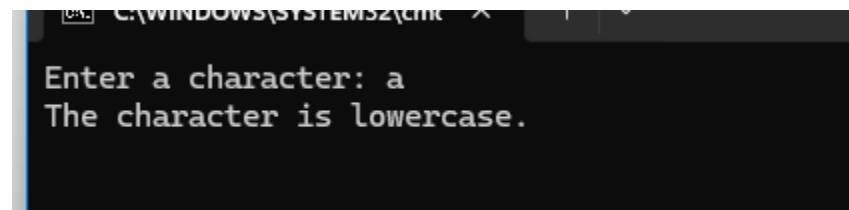
```
import java.util.*;

class Example30 {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

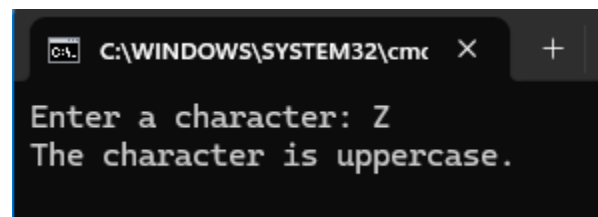
        System.out.print("Enter a character: ");

        char ch = scanner.next().charAt(0);
        if (ch >= 'a' && ch <= 'z') {
            System.out.println("The character is lowercase.");
        } else if (ch >= 'A' && ch <= 'Z') {
            System.out.println("The character is uppercase.");
        } else {
            System.out.println("It is not an English letter.");
        }
    }
}
```

Outputs.

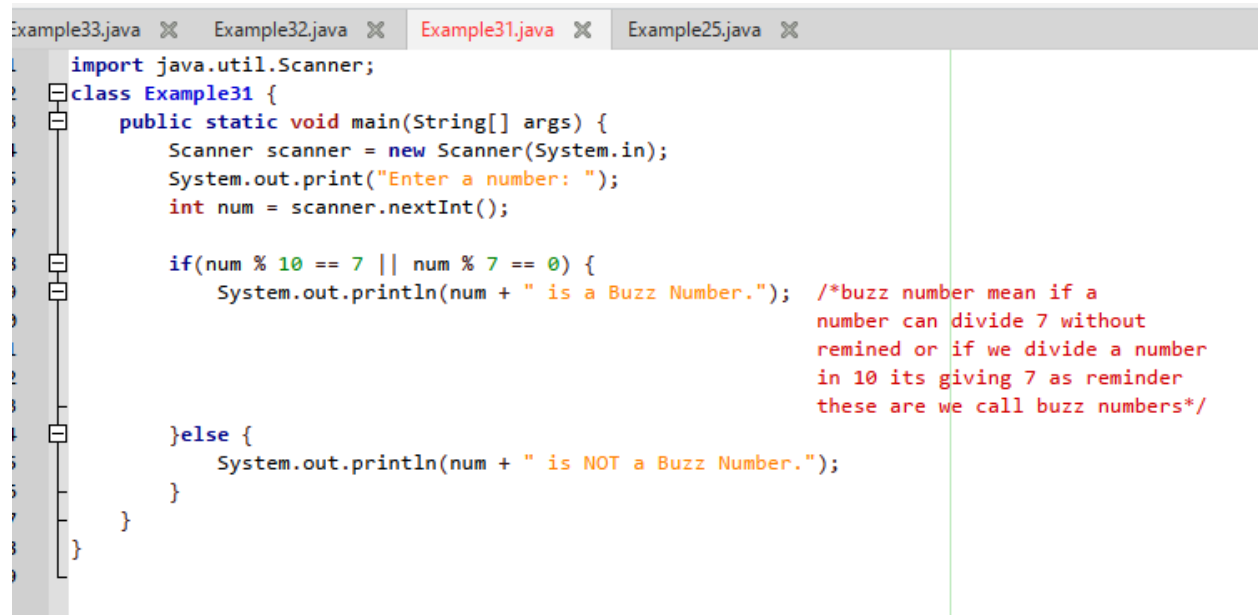


```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter a character: a
The character is lowercase.
```



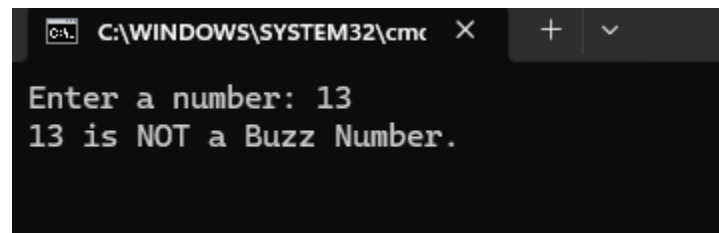
```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter a character: Z
The character is uppercase.
```

Q31.

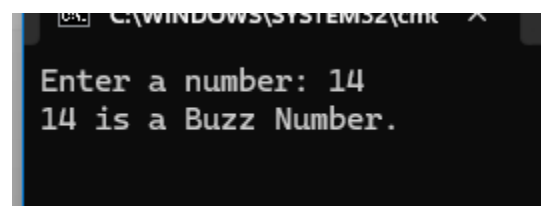


```
Example33.java X Example32.java X Example31.java X Example25.java X
1 import java.util.Scanner;
2 class Example31 {
3     public static void main(String[] args) {
4         Scanner scanner = new Scanner(System.in);
5         System.out.print("Enter a number: ");
6         int num = scanner.nextInt();
7
8         if(num % 10 == 7 || num % 7 == 0) {
9             System.out.println(num + " is a Buzz Number."); /*buzz number mean if a
10                                                            number can divide 7 without
11                                                            remined or if we divide a number
12                                                            in 10 its giving 7 as reminder
13                                                            these are we call buzz numbers*/
14         }else {
15             System.out.println(num + " is NOT a Buzz Number.");
16         }
17     }
18 }
```

Outputs.



```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter a number: 13
13 is NOT a Buzz Number.
```



```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter a number: 14
14 is a Buzz Number.
```

Q32.

```
ample33.java ✕ Example32.java ✕ Example25.java ✕
import java.util.*;
class Example32{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the number of classes held : ");
        int hClasses = input.nextInt();

        System.out.print("Enter the number of attend classes : ");
        int aClasses = input.nextInt();

        double aPer = ((double)aClasses/hClasses)*100;
        if(aPer>=70){
            System.out.print("eligibility for the PRF exam");
        }else{
            System.out.print("Do you have a Medical enter 'Y' or 'N' : ");
            char med = input.next().charAt(0);

            if(med=='Y' || med=='y'){
                System.out.print("you can sit for the PRF exam");
            }else if(med=='N' || med=='n'){
                System.out.print("you can't sit for the PRF exam");
            }
        }
    }
}
```

Outputs.

```
Enter the number of classes held : 100
Enter the number of attend classes : 90
eligibility for the PRF exam
-----
```

```
Enter the number of classes held : 100
Enter the number of attend classes : 34
Do you have a Medical enter 'Y' or 'N' : y
you can sit for the PRF exam
-----
```

```
Enter the number of classes held : 100
Enter the number of attend classes : 35
Do you have a Medical enter 'Y' or 'N' : n
you can't sit for the PRF exam
-----
```

Q33.

```
Example33.java Example33.java
import java.util.*;
class Example33{
    public static void main (String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Is the permanent employee? (yes/no): ");
        String pEmp = input.next().toLowerCase();

        if (pEmp.equals("yes")) {
            System.out.print("Enter base salary: ");
            double salary = input.nextDouble();

            System.out.print("Enter years of service: ");
            double yearSer = input.nextDouble();

            double bPercentage = 0;
            if (yearSer < 5) {
                bPercentage = 0.10;
            } else if (yearSer <= 10) {
                bPercentage = 0.15;
            } else {
                bPercentage = 0.25;
            }

            double bonusAmount = salary * bPercentage;
            double totalSalary = salary + bonusAmount;

            System.out.println("Base Salary: "+ salary);
            System.out.println("Bonus: "+ bonusAmount);
            System.out.println("totalSalary "+ totalSalary);
        }else{
            System.out.print("Enter base salary: ");
            double salary = input.nextDouble();
            System.out.println("Base Salary: "+ salary);
        }
    }
}
```

Outputs.

```
Is the employee permanent? (yes/no): no
Enter base salary: 12000
Base Salary: 12000.0
```

```
Is the employee permanent? (yes/no): yes
Enter base salary: 490000
Enter years of service: 4
Base Salary: 490000.0
Bonus: 49000.0
totalSalary 539000.0
```

```
Is the employee permanent? (yes/no): yes
Enter base salary: 490000
Enter years of service: 6
Base Salary: 490000.0
Bonus: 73500.0
totalSalary 563500.0
```

```
Is the employee permanent? (yes/no): yes
Enter base salary: 490000
Enter years of service: 11
Base Salary: 490000.0
Bonus: 122500.0
totalSalary 612500.0
```

Q34.

```
import java.util.*;
class Example34{
    public static void main (String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("How many books did you buying : ");
        int nBook = input.nextInt();

        final int bPrice =100;
        double Subtotal = nBook * bPrice;
        if (Subtotal>5000){
            double dis = Subtotal*0.10;
            double tot = Subtotal - dis;
            System.out.printf("Subtotal : %9.2f \n",Subtotal);
            System.out.printf("Discount : %9.2f \n",dis);
            System.out.printf("TOTAL      : %9.2f \n",tot);

        }else{
            double tot = Subtotal ;
            String dis = "-";
            System.out.printf("Subtotal : %9.2f \n",Subtotal);
            System.out.printf("Discount : %9.2s \n",dis);
            System.out.printf("TOTAL      : %9.2f \n",tot);
        }
    }
}
```

Outputs.

```
How many books did you buying : 34
Subtotal :   3400.00
Discount :      -
TOTAL      :   3400.00

-----
```

```
How many books did you buying : 69
Subtotal :   6900.00
Discount :    690.00
TOTAL      :   6210.00

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q35.

```
Example25.java X Example33.java X Example34.java X Example35.java X
import java.util.*;
class Example35{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter today temperature : ");
        double temp = input.nextDouble();

        if(temp>=80){
            System.out.print("The appropriate activity is Swimming");
        }else if(80>temp && temp>=60){
            System.out.print("The appropriate activity is Tennis");
        }else if(60>temp && temp>=40){
            System.out.print("The appropriate activity is Golf");
        }else{
            System.out.print("The appropriate activity is Skiing");
        }
    }
}
```

Outputs.

```
Enter today temperature : 81
The appropriate activity is Swimming
-----
(program exited with code: 0)
```

```
Enter today temperature : 76
The appropriate activity is Tennis
-----
```

```
Enter today temperature : 59
The appropriate activity is Golf
-----
```

```
Enter today temperature : 37
The appropriate activity is Skiing
-----
```

Q36.

```
import java.util.*;
class Example36 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter an alphabet letter: ");
        char ch = input.next().toUpperCase().charAt(0);

        if (ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U') {
            System.out.println(ch + " is a Vowel.");
        } else if (ch >= 'A' && ch <= 'Z') {
            System.out.println(ch + " is a Consonant.");
        } else {
            System.out.println("Invalid input! Please enter an alphabet letter.");
        }
    }
}
```

Outputs.

```
Enter an alphabet letter: f
F is a Consonant.

-----
(program exited with code: 0)
Press any key to continue . . . |
```

```
Enter an alphabet letter: i
I is a Vowel.
```

```
Enter an alphabet letter: 1
Invalid input! Please enter an alphabet letter.
```

Q37.

```
import java.util.*;
class Example37{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter current annual salary: ");
        double cSalary = input.nextDouble();

        System.out.print("Enter performance rating (1 =excellent, 2 =good, 3 =poor): ");
        int rating = input.nextInt();

        double rRate = 0; //raiseRate

        if (rating == 1) {
            rRate = 0.06;
        } else if (rating == 2) {
            rRate = 0.04;
        } else if (rating == 3) {
            rRate = 0.015;
        } else {
            System.out.println("Invalid rating entered!");
        }

        double rAmount = cSalary * rRate;
        double newSalary = cSalary + rAmount;
        System.out.printf("New Salary      : %.2f\n", newSalary);
    }
}
```

Outputs.

```
Enter current annual salary: 120000
Enter performance rating (1 =excellent, 2 =good, 3 =poor): 1
New Salary      : 127200.00
```

```
Enter current annual salary: 120000
Enter performance rating (1 =excellent, 2 =good, 3 =poor): 2
New Salary      : 124800.00
```

```
Enter current annual salary: 120000
Enter performance rating (1 =excellent, 2 =good, 3 =poor): 3
New Salary      : 121800.00
```

Q38.

```
Example25.java Example33.java Example34.java Example35.java Example36.java Example37.java
import java.util.*;
class Example38{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter your attendance percentage : ");
        double aPer = input.nextDouble();

        System.out.print("Enter your Final test marks : ");
        double fMarks = input.nextDouble();

        if (aPer > 80 && fMarks > 50) {
            System.out.println("Eligible to sit for the O/L examination.");
        } else {
            System.out.println("Not eligible to sit for the O/L examination.");
        }
    }
}
```

Outputs.

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter your attendance percentage : 91
Enter your Final test marks : 69
Eligible to sit for the O/L examination.
=====
```

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter your attendance percentage : 33
Enter your Final test marks : 60
Not eligible to sit for the O/L examination.
```

```
Enter your attendance percentage : 97
Enter your Final test marks : 22
Not eligible to sit for the O/L examination.
```

Q39.

```
Example39.java Example40.java Example41.java Example42.java Example43.java
import java.util.*;
class Example39 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter time in 24-hour format (e.g. 8.30 ): ");
        double time = input.nextDouble();

        if(time%1<0 || time%1>=60){
            System.out.println("Invalid time entered!");
        }else if (time >= 0.00 && time < 12.00) {
            System.out.println("Good morning");
        }else if(time >= 12.00 && time < 16.00) {
            System.out.println("Good afternoon");
        }else if(time >= 16.00 && time < 19.00) {
            System.out.println("Good evening");
        }else {
            System.out.println("Good night");
        }
    }
}
```

Outputs.

```
C:\WINDOWS\SYSTEM32\cmd
Enter time in 24-hour format (e.g. 8.30 ): 4.40
Good morning
```

```
C:\WINDOWS\SYSTEM32\cmd
Enter time in 24-hour format (e.g. 8.30 ): 13.50
Good afternoon

-----
(program exited with code: 0)
```

Q40.

```
Example40.java  Example41.java  Example42.java  Example43.java  Example44.java  Example45.java  Example46.java
import java.util.*;
class Example40{
    public static void main (String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a Year : ");
        int year = input.nextInt();
        input.nextLine();

        System.out.print("Enter a month : ");
        String month = input.nextLine().toLowerCase();

        int days = 0;

        boolean leapYear = false;
        if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 100)){
            leapYear = true;
            System.out.println("This is a leap year");
        }else{
            System.out.println("This is not a leap year");
        }

        if(month.equals("january") || month.equals("march") || month.equals("may") || month.equals("july") ||
            month.equals("august") || month.equals("october") || month.equals("december") ){

            days = 31;

        }else if(month.equals("april") || month.equals("june") || month.equals("september") || month.equals("november")){
            days = 30;
        }else if(month.equals("february")){
            if(leapYear){
                days = 29;
            }else {
                days = 28;
            }
        }else{
            System.out.println("Invalid month name entered!");
        }

        System.out.println("In "+year+" "+month+" has "+days+" days.");
    }
}
```

Output.

```
Enter a Year : 2004
Enter a month : march
This is a leap year
In 2004 march has 31 days.
```

```
import java.util.*;
class Example41{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the number of copies that you want : ");
        int nCopy = input.nextInt();

        double perCopy = 0;
        double tot = 0;
        if(0<=nCopy && nCopy<100){
            perCopy = 30;
            tot = perCopy*nCopy;
        }else if(100<=nCopy && nCopy<500){
            perCopy = 28;
            tot = perCopy*nCopy;
        }else if(500<=nCopy && nCopy<800){
            perCopy = 27;
            tot = perCopy*nCopy;
        }else if(800<=nCopy && nCopy<1000){
            perCopy = 26;
            tot = perCopy*nCopy;
        }else if(1000<=nCopy){
            perCopy = 25;
            tot = perCopy*nCopy;
        }else{
            System.out.print("Invalid number of copies");
        }

        System.out.println("price per copy "+perCopy);
        System.out.println("Total price : "+tot);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
Enter the number of copies that you want : 204
price per copy 28.0
Total price : 5712.0
```

Q42.

```
Example42.java X Example43.java X Example44.java X Example45.java X Example46.java
import java.util.*;
class Example42 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter your waist size in inches(28-42): ");
        int wSize = input.nextInt();
        if (wSize >= 28 && wSize <= 29) {
            System.out.println("Size: X-small");
        } else if (wSize >= 30 && wSize <= 31) {
            System.out.println("Size: Small ");
        } else if (wSize >= 32 && wSize <= 34) {
            System.out.println("Size: Medium");
        } else if (wSize >= 36 && wSize <= 38) {
            System.out.println("Size: Large");
        } else if (wSize >= 40 && wSize <= 42) {
            System.out.println("Size: X-Large");
        } else {
            System.out.println("Invalid or unavailable size!");
        }
    }
}
```

Outputs.

```
Enter your waist size in inches(28-42): 36
Size: Large
```

```
Enter your waist size in inches(28-42): 29
Size: X-small
```

Q43.

```
Example43.java  Example44.java  Example45.java  Example46.java  Example47.java
import java.util.Scanner;
class Example43{
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the first number : ");
        double num1 = input.nextDouble();

        System.out.print("Enter the second number : ");
        double num2 = input.nextDouble();

        System.out.print("Enter the operator (+,-,*,/,%,^): ");
        char operator = input.next().charAt(0);

        double result = 0;
        boolean validOperation = true;

        switch (operator) {
            case '+': result = num1 + num2;
                     break;
            case '-': result = num1 - num2;
                     break;
            case '*': result = num1 * num2;
                     break;
            case '/':
                if (num2 != 0) {
                    result = num1 / num2;
                } else {
                    System.out.println("Error: Cannot divide by zero!");
                    validOperation = false;
                }
                break;
            case '%': result = num1 % num2;
                     break;
            case '^': result = Math.pow(num1, num2);
                     break;
            default: System.out.println("Invalid operator!");
                    validOperation = false;
        }
        System.out.println(num1+" "+operator+" "+num2+" = "+result);
    }
}
```

Outputs.

```
Enter the first number : 23
Enter the second number : 34
Enter the operator (+,-,*,/,%,^): *
23.0 * 34.0 = 782.0
```

```
Enter the first number : 12
Enter the second number : 54
Enter the operator (+,-,*,/,%,^): ^
12.0 ^ 54.0 = 1.887066854784446E58
```

Q44.

```
Example44.java Example43.java Example40.java Example41.java Example40.java Example43
1 import java.util.*;
2 class Example44 {
3     public static void main(String[] args) {
4         Scanner input = new Scanner(System.in);
5         System.out.print("Enter a day number for February 2022 (1 - 28): ");
6         int day = input.nextInt();
7
8         if (day >= 1 && day <= 28) {
9             int dayName = day % 7;
10
11             switch (dayName) {
12                 case 1: System.out.println("Tuesday");
13                     break;
14                 case 2: System.out.println("Wednesday");
15                     break;
16                 case 3: System.out.println("Thursday");
17                     break;
18                 case 4: System.out.println("Friday");
19                     break;
20                 case 5: System.out.println("Saturday");
21                     break;
22                 case 6: System.out.println("Sunday");
23                     break;
24                 case 0: System.out.println("Monday");
25                     break;
26             }
27         }
28         else{
29             System.out.println("Invalid day! Please enter a number between 1 and 28.");
30         }
31     }
32 }
33 }
```

Outputs.

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter a day number for February 2022 (1 - 28): 4
Friday
```

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter a day number for February 2022 (1 - 28): 7
Monday
```

Q45.

```
xample45.java  Example46.java  Example47.java  Example48.java  Example49.java  Example50.java  Exam
import java.util.*;
class Example45{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the Row number : ");
        int row = input.nextInt();

        System.out.print("Enter the column number : ");
        int col = input.nextInt();

        if((row>0 && row<9) || (col>0 && col<9)){
            if(row%2==0){
                if(col%2==0){
                    System.out.print("You are Entered cell color is white");
                }else{
                    System.out.print("You are Entered cell color is black");
                }
            }else{
                if(col%2==1){
                    System.out.print("You are Entered cell color is white");
                }else{
                    System.out.print("You are Entered cell color is black");
                }
            }
        }else{
            System.out.print("Error ,enter a valid raw(1-8) or colum(1-8) number");
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
Enter the Row number : 5
Enter the column number : 7
You are Entered cell color is white
_____
```

```
C:\WINDOWS\SYSTEM32\cmd
Enter the Row number : 8
Enter the column number : 1
You are Entered cell color is black
```

Q46.

```
import java.util.*;
class Example46 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter first number: ");
        int a = input.nextInt();

        System.out.print("Enter second number: ");
        int b = input.nextInt();

        System.out.print("Enter third number: ");
        int c = input.nextInt();

        int a2 = a * a;
        int b2 = b * b;
        int c2 = c * c;

        if ((a2 + b2 == c2) || (a2 + c2 == b2) || (b2 + c2 == a2)) {
            System.out.println("The numbers form a Pythagorean triple.");
        } else {
            System.out.println("The numbers DO NOT form a Pythagorean triple.");
        }
    }
}
```

Outputs.

```
Enter first number: 2
Enter second number: 5
Enter third number: 6
The numbers DO NOT form a Pythagorean triple.
```

```
Enter first number: 3
Enter second number: 4
Enter third number: 5
The numbers form a Pythagorean triple.
```

Q47.

```
import java.util.*;
class Example47 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter month (1-12): ");
        int month = input.nextInt();

        System.out.print("Enter day (1-31): ");
        int day = input.nextInt();

        String season = "";

        if ((month==12 && day>=21) || month==1 || month==2 || (month==3 && day<=19)) {
            season = "Winter";
        }
        else if ((month==3 && day>=20) || month==4 || month==5 || (month==6 && day<=20)) {
            season = "Spring";
        }
        else if ((month==6 && day>=21) || month==7 || month==8 || (month==9 && day<=21)) {
            season = "Summer";
        }
        else if ((month==9 && day>=22) || month==10 || month==11 || (month==12 && day<=20)) {
            season = "Autumn";
        }
        else {
            season = "Invalid Date";
        }
        System.out.println("Season: " + season);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd X
Enter month (1-12): 12
Enter day (1-31): 4
Season: Autumn
```

```
1 Enter month (1-12): 6
Enter day (1-31): 10
Season: Spring
```

Q48.

```
import java.util.Scanner;
class Example48 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter month (1-12): ");
        int month = input.nextInt();

        System.out.print("Enter day (1-31): ");
        int day = input.nextInt();

        String sign = "";
        if ((month == 12 && day >= 22) || (month == 1 && day <= 19)) {
            sign = "Capricornus";
        }
        else if ((month == 1 && day >= 20) || (month == 2 && day <= 18)) {
            sign = "Aquarius";
        }
        else if ((month == 2 && day >= 19) || (month == 3 && day <= 20)) {
            sign = "Pisces";
        }
        else if ((month == 3 && day >= 21) || (month == 4 && day <= 19)) {
            sign = "Aries (Ram)";
        }
        else if ((month == 4 && day >= 20) || (month == 5 && day <= 20)) {
            sign = "Taurus";
        }
        else if ((month == 5 && day >= 21) || (month == 6 && day <= 21)) {
            sign = "Gemini";
        }
        else if ((month == 6 && day >= 22) || (month == 7 && day <= 22)) {
            sign = "Cancer";
        }
        else if ((month == 7 && day >= 23) || (month == 8 && day <= 22)) {
            sign = "Leo";
        }
        else if ((month == 8 && day >= 23) || (month == 9 && day <= 22)) {
            sign = "Virgo";
        }
        else if ((month == 9 && day >= 23) || (month == 10 && day <= 23)) {
            sign = "Libra";
        }
        else if ((month == 10 && day >= 24) || (month == 11 && day <= 21)) {
            sign = "Scorpius";
        }
        else if ((month == 11 && day >= 22) || (month == 12 && day <= 21)) {
            sign = "Sagittarius";
        }
        else {
            sign = "Invalid Date";
        }
        System.out.println("Astrological Sign: " + sign);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter month (1-12): 12
Enter day (1-31): 4
Astrological Sign: Sagittarius
```

```
Enter month (1-12): 8
Enter day (1-31): 21
Astrological Sign: Leo
```

Q49.

```
Example49.java X Example50.java X Example51.java X Example52.java X Examp
import java.util.*;
class Example49 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter Basic Salary: ");
        double basicSalary = input.nextDouble();

        double ha = 0;
        double ta = 0;
        if(0 < basicSalary){
            if( basicSalary <= 10000){
                ha = basicSalary * 0.20;
                ta = basicSalary * 0.60;
            }else if(basicSalary <= 20000){
                ha = basicSalary * 0.25;
                ta = basicSalary * 0.70;
            }else{
                ha = basicSalary * 0.30;
                ta = basicSalary * 0.75;
            }
            double grossSalary = basicSalary + ha + ta;
            System.out.println("Gross Salary: " + grossSalary);
        }else{
            System.out.print("Invalid Salary.");
        }
    }
}
```

Output.

```
Enter Basic Salary: 34000
Gross Salary: 69700.0
```

```
s
s Enter Basic Salary: 120000
d Gross Salary: 246000.0
d
```

Q50.

```
Example50.java  Example51.java  Example52.java  Example53.java  Example54.j
1  import java.util.*;
2  class Example50{
3      public static void main(String[] args){
4          Scanner input = new Scanner(System.in);
5          System.out.print("Enter angle a: ");
6          int a = input.nextInt();
7
8          System.out.print("Enter angle b: ");
9          int b = input.nextInt();
10
11         System.out.print("Enter angle c: ");
12         int c = input.nextInt();
13         if(a > 0 && b > 0 && c > 0 && (a + b + c == 180)){
14             System.out.println("The triangle is valid.");
15         }else{
16             System.out.println("The triangle is NOT valid.");
17         }
18     }
19 }
20 }
```

Output.

```
Enter angle a: 34
Enter angle b: 56
Enter angle c: 23
The triangle is NOT valid.
```

```
Enter angle a: 60
Enter angle b: 60
Enter angle c: 60
The triangle is valid.
```

Q51.

```
Example51.java X Example52.java X Example53.java X
import java.util.*;
class Example51 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter age: ");
        int age = input.nextInt();

        if (age > 65) {
            System.out.println("Senior");
        } else if (age > 20) {
            System.out.println("Adult");
        } else if (age > 13) {
            System.out.println("Teenager");
        } else if (age > 1) {
            System.out.println("Child");
        } else {
            System.out.println("Infant");
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd X
Enter age: 45
Adult
```

```
C:\WINDOWS\SYSTEM32\cmd X
Enter age: 15
Teenager
-----
(program exited with code: 0)
```

Q52.

```
xample52.java  Example53.java  Example54.java  Example55.java  Exampl
import java.util.*;
class Example52{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the X coordinate : ");
        double x = input.nextDouble();

        System.out.print("Enter the Y coordinate : ");
        double y = input.nextDouble();

        if(y>0){
            if(x>0){
                System.out.print("Quadrant I");
            }else{
                System.out.print("Quadrant II");
            }
        }else if(y<0){
            if(x>0){
                System.out.print("Quadrant IV");
            }else{
                System.out.print("Quadrant III");
            }
        }else{
            System.out.print("it is (0,0)");
        }
    }
}
```

Output.

```
Enter the X coordinate : 3
Enter the Y coordinate : 7
Quadrant I
_____
```

```
C:\WINDOWS\SYSTEM32\cmd  +  v
Enter the X coordinate : 4
Enter the Y coordinate : -67
Quadrant IV
_____
```

Q53.

```
Example53.java Example54.java Example55.java Example56.java Example57.java
import java.util.*;
class Example53{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter Systolic Blood Pressure (SBP): ");
        int sbp = input.nextInt();

        System.out.print("Enter Diastolic Blood Pressure (DBP): ");
        int dbp = input.nextInt();

        if (sbp >= 130 || dbp >= 90) {
            System.out.println("Status: High Pressure");
        } else if (sbp <= 100 || dbp <= 70) {
            System.out.println("Status: Low Pressure");
        } else {
            System.out.println("Status: Normal");
        }
    }
}
```

Output.

```
Enter Systolic Blood Pressure (SBP): 131
Enter Diastolic Blood Pressure (DBP): 43
Status: High Pressure
```

```
Enter Systolic Blood Pressure (SBP): 90
Enter Diastolic Blood Pressure (DBP): 43
Status: Low Pressure
```

Q54.

```
Example54.java Example54.java Example54.java Example54.java Example54.java
import java.util.*;
class Example54{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter Website URL: ");
        String url = input.nextLine();

        if(url.endsWith(".com")){
            System.out.println("Commercial Website");
        }else if(url.endsWith(".org")){
            System.out.println("Organization Website");
        }else if(url.endsWith(".lk")){
            System.out.println("Sri Lankan Website");
        }else{
            System.out.println("Unknown Website Type");
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
Enter Website URL: .com
Commercial Website

-----
(program exited with code: 0)
```

```
C:\WINDOWS\SYSTEM32\cmd
Enter Website URL: .lk
Sri Lankan Website

-----
```

Q55

```
Example55.java X Example56.java X Example57.java X Example58.java X Example59.java X
import java.util.Scanner;
class Example55{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter the number: ");
        int num = input.nextInt();

        System.out.print("Enter lower bound: ");
        int lower = input.nextInt();

        System.out.print("Enter upper bound: ");
        int upper = input.nextInt();

        if (num >= lower && num <= upper) {
            System.out.println("The number is within the range.");
        } else {
            System.out.println("The number is NOT within the range.");
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter the number: 34
Enter lower bound: 1
Enter upper bound: 100
The number is within the range.
```

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter the number: 100
Enter lower bound: 1
Enter upper bound: 99
The number is NOT within the range.
```

Q56.

56. Rewrite in Java the following statement without using the NOT operator (!):

`item = !( (a<10) || (b>=50) )`

Example 56: Rewrite the following statement without using the NOT operator (!)

Output.

ample56.java ✕ Example57.java ✕ Example58.java ✕

```
boolean item = (a >= 10) && (b < 50);
```

Q57.

```
1 import java.util.*;
2 public class Example57 {
3     public static void main(String[] args) {
4         Scanner input = new Scanner(System.in);
5         System.out.print("Enter First Date (month): ");
6         int m1 = input.nextInt();
7         System.out.print("Enter First Date (day): ");
8         int d1 = input.nextInt();
9         System.out.print("Enter First Date (year): ");
10        int y1 = input.nextInt();
11
12        System.out.println();
13
14        System.out.print("Enter Second Date (month): ");
15        int m2 = input.nextInt();
16        System.out.print("Enter Second Date (day): ");
17        int d2 = input.nextInt();
18        System.out.print("Enter Second Date (year): ");
19        int y2 = input.nextInt();
20
21        System.out.println();
22
23        if(m1 < 1 || m1 > 12 || d1 < 1 || d1 > 31 || y1 <= 0 ||
24           m2 < 1 || m2 > 12 || d2 < 1 || d2 > 31 || y2 <= 0){
25            System.out.println("Invalid Date!");
26        }else if (y1 < y2 || (y1 == y2 && m1 < m2) || (y1 == y2 && m1 == m2 && d1 < d2)){
27            System.out.println("First date comes first.");
28        }else if (y1 > y2 || (y1 == y2 && m1 > m2) || (y1 == y2 && m1 == m2 && d1 > d2)){
29            System.out.println("Second date comes first.");
30        }else{
31            System.out.println("Both dates are the same.");
32        }
33    }
34 }
```

Output.

```
Enter First Date (month): 12
Enter First Date (day): 3
Enter First Date (year): 2005

Enter Second Date (month): 12
Enter Second Date (day): 13
Enter Second Date (year): 2005

First date comes first.
```

```
Enter First Date (month): 12
Enter First Date (day): 2
Enter First Date (year): 2005

Enter Second Date (month): 4
Enter Second Date (day): 4
Enter Second Date (year): 2005

Second date comes first.
```

Q58.

```
Example58.java Example59.java Example60.java
import java.util.Scanner;
class Example58 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter year: ");
        int y = input.nextInt();

        System.out.print("Enter month: ");
        int m = input.nextInt();

        System.out.print("Enter day: ");
        int d = input.nextInt();

        int y0 = y - (14 - m) / 12;
        int x = y0 + y0 / 4 - y0 / 100 + y0 / 400;
        int m0 = m + 12 * ((14 - m) / 12) - 2;
        int d0 = (d + x + (31 * m0) / 12) % 7;

        String dayName = "";

        if(d0 == 0){
            dayName = "Sunday";
        }else if (d0 == 1){
            dayName = "Monday";
        }else if (d0 == 2){
            dayName = "Tuesday";
        }else if (d0 == 3){
            dayName = "Wednesday";
        }else if (d0 == 4){
            dayName = "Thursday";
        }else if (d0 == 5){
            dayName = "Friday";
        }else if (d0 == 6){
            dayName = "Saturday";
        }
        System.out.println("Day of the week: " + dayName);
    }
}
```

Outputs.

```
Enter year: 2005
Enter month: 2
Enter day: 20
Day of the week: Sunday
```

```
Enter year: 2004
Enter month: 06
Enter day: 10
Day of the week: Thursday
```

Q59.

```
import java.util.Scanner;
class Example59 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter R (0-255): ");
        double r = input.nextDouble();

        System.out.print("Enter G (0-255): ");
        double g = input.nextDouble();

        System.out.print("Enter B (0-255): ");
        double b = input.nextDouble();

        double c, m, y, k;

        if(r == 0 && g == 0 && b == 0){
            c = 0.0;
            m = 0.0;
            y = 0.0;
            k = 1.0;
        }else{
            double rPrime = r / 255.0;
            double gPrime = g / 255.0;
            double bPrime = b / 255.0;
            double w = Math.max(rPrime, Math.max(gPrime, bPrime));
            c = (w - rPrime) / w;
            m = (w - gPrime) / w;
            y = (w - bPrime) / w;
            k = 1.0 - w;
        }
        System.out.println("Cyan (C)   = " + c);
        System.out.println("Magenta (M) = " + m);
        System.out.println("Yellow (Y) = " + y);
        System.out.println("Black (K)   = " + k);
    }
}
```

Output.

```
Enter R (0-255): 34
Enter G (0-255): 25
Enter B (0-255): 46
Cyan (C)   = 0.2608695652173913
Magenta (M) = 0.45652173913043476
Yellow (Y)  = 0.0
Black (K)   = 0.8196078431372549
```

Q60.

```
xample60.java X
import java.util.Scanner;

public class Example60 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter parcel weight (in kg): ");
        double weight = input.nextDouble();

        double charge;

        if (weight <= 5) {
            charge = 500;
        } else {
            double extraWeight = weight - 5;
            charge = 500 + (extraWeight * 100);
        }

        System.out.println("Courier Charge = Rs. " + charge);

        input.close();
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter parcel weight (in kg): 23
Courier Charge = Rs. 2300.0
```

